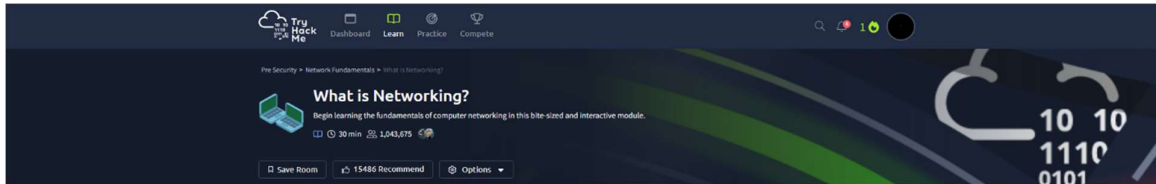


Day 02: What is Networking?

Link: <https://tryhackme.com/room/whatisnetworking>



Task 01: What is Networking?

Networks are simply things connected. For example, your friendship circle: you are all connected because of similar interests, hobbies, skills and sorts.

Networks can be found in all walks of life:

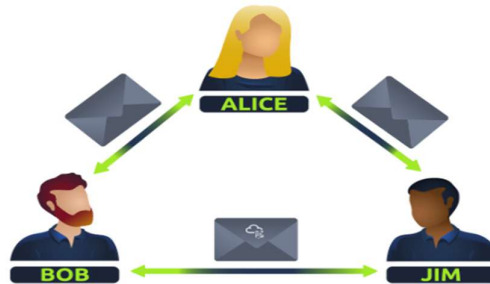
- A city's public transportation system
- Infrastructure such as the national power grid for electricity
- Meeting and greeting your neighbours
- Postal systems for sending letters and parcels

But more specifically, in computing, networking is the same idea, just dispersed to technological devices. Take your phone as an example; the reason that you have it is to access things. We'll cover how these devices communicate with each other and the rules that follow.

In computing, a network can be formed by anywhere from 2 devices to billions. These devices include everything from your laptop and phone to security cameras, traffic lights and even farming!

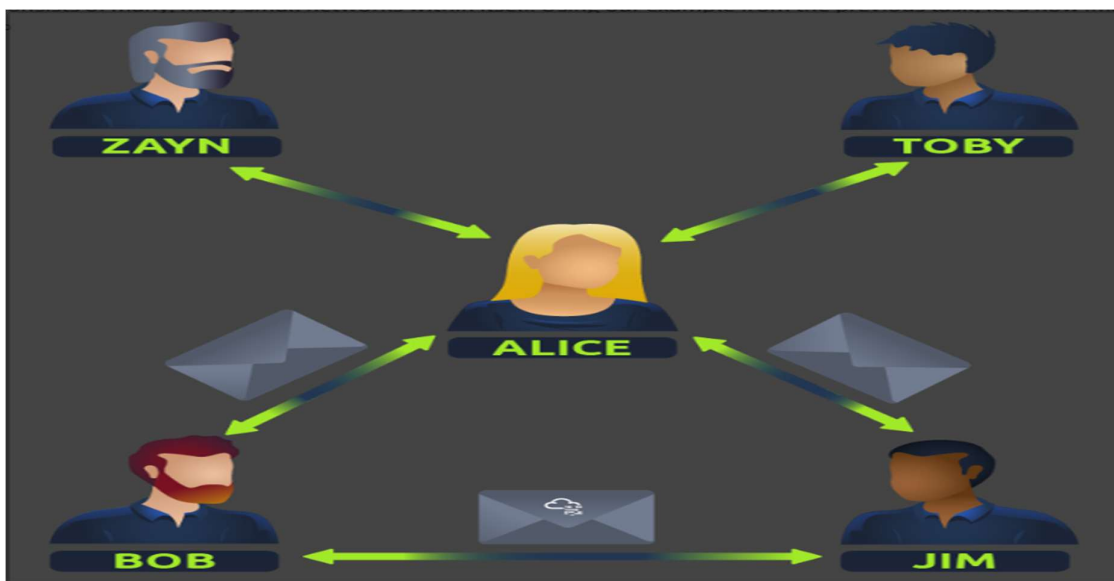
Networks are integrated into our everyday life. Be it gathering data for the weather, delivering electricity to homes or even determining who has the right of way at a road. Because networks are so embedded in the modern-day, networking is an essential concept to grasp in cybersecurity.

Take the diagram below as an example, Alice, Bob and Jim have formed their network! We'll come onto this a bit later on.



Networks come in all shapes and sizes, which is something that we will also come on to discuss throughout this module.

Task 02: What is the Internet?



Task 03: Identifying Devices on a Network

IP Addresses

Briefly, an IP address (or Internet Protocol) address can be used as a way of identifying a host on a network for a period of time, where that IP address can then be associated with another device without the IP address changing. First, let's split up precisely what an IP address is in the diagram below:



An IP address is a set of numbers that are divided into four octets. The value of each octet will summarise to be the IP address of the device on the network. This number is calculated through a technique known as IP addressing & subnetting, but that is for another day. What's important to understand here is that IP addresses can change from device to device but cannot be active simultaneously more than once within the same network.

IP Addresses follow a set of standards known as protocols. These protocols are the backbone of networking and force many devices to communicate in the same language, which is something that we'll come onto another time. However, we should recall that devices can be on both a private and public network. Depending on where they are will determine what type of IP address they have: a public or private IP address.

A public address is used to identify the device on the Internet, whereas a private address is used to identify a device amongst other devices. Take the table & screenshot below as an example. Here we have two devices on a private network:

Device Name	IP Address	IP Address Type
DESKTOP-KJES7FD	192.168.1.77	Private
DESKTOP-KJES7FD	86.157.52.21	Public
CMNatic-PC	192.168.1.74	Private
CMNatic-PC	86.157.52.21	Public

 DESKTOP-KJES7FD 5 GHz	IP address: 192.168.1.77 (DHCP) MAC address: EC:5C:68:C3:7E:51
 CMNatic-PC 5 GHz	IP address: 192.168.1.74 (DHCP) MAC address: 50:3E:AA:E8:3B:64

These two devices will be able to use their private IP addresses to communicate with each other. However, any data sent to the Internet from either of these devices will be identified by the same public IP address. Public IP addresses are given by your Internet Service Provider (or ISP) at a monthly fee (your bill):

My Public IPv4 is: 86.157.52.21

Location: ENG GB

ISP: Secure Communications TLD

As more and more devices become connected, it is becoming increasingly harder to get a public address that isn't already in use. For example, Cisco, an industry giant in the world of networking, estimated that there would be approximately 50 billion devices connected on the Internet by the end of 2021. (Cisco, 2021). Enter IP address versions. So far, we have only discussed one version of the Internet Protocol addressing scheme known as IPv4, which uses a numbering system of 2³² IP addresses (4.29 billion) -- so you can see why there is such a shortage!

IPv6 is a new iteration of the Internet Protocol addressing scheme to help tackle this issue. Although it is seemingly more daunting, it boasts a few benefits:

- Supports up to 2¹²⁸ of IP addresses (340 trillion-plus), resolving the issues faced with IPv4
- More efficient due to new methodologies

The screenshot below compares both an IPv6 and IPv4 address.

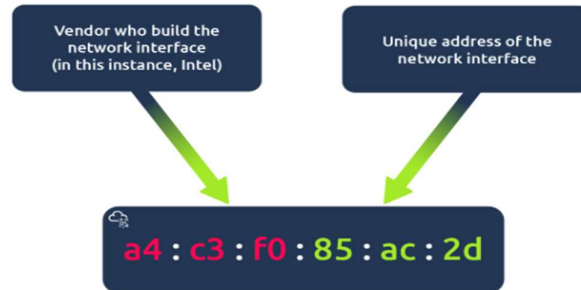
My IP Address is:

IPv6: **2a00:22c4:a531:c500:425f:ccc6:c36b:f64d**

IPv4: **86.157.52.21**

MAC Addresses

Devices on a network will all have a physical network interface, which is a microchip board found on the device's motherboard. This network interface is assigned a unique address at the factory it was built at, called a **MAC (Media Access Control)** address. The MAC address is a **twelve-character** hexadecimal number (a base sixteen numbering system used in computing to represent numbers) split into two's and separated by a colon. These colons are considered separators. For example, `a4:c3:f0:85:ac:2d`. The first six characters represent the company that made the network interface, and the last six is a unique number.



However, an interesting thing with MAC addresses is that they can be faked or "spoofed" in a process known as spoofing. This spoofing occurs when a networked device pretends to identify as another using its MAC address. When this occurs, it can often break poorly implemented security designs that assume that devices talking on a network are trustworthy. Take the following scenario: A **firewall** is configured to allow any communication going to and from the MAC address of the administrator. If a device were to pretend or "spoof" this MAC address, the **firewall** would now think that it is receiving communication from the administrator when it isn't.

Places such as cafes, coffee shops, and hotels alike often use MAC address control when using their "Guest" or "Public" Wi-Fi. This configuration could offer better services, i.e. a faster connection for a price if you are willing to pay the fee per device. The **interactive lab attached to this task** has been made to replicate this scenario!

Practical

The interactive labs simulate a hotel Wi-Fi network where you have to pay for the service. You'll note that the router is not allowing Bob's packets (blue) to the TryHackMe website and is placing them in the bin, but Alice's packets (green) are going through fine because she has paid for Wi-Fi. Try changing Bob's MAC address to the same as Alice's to see what happens.

Deploy the interactive lab and proceed to answer the following questions below.

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Deploy the interactive lab and proceed to answer the following questions below.

Answer the questions below

What does the term "IP" stand for?

Internet Protocol ✓ Correct Answer

What is each section of an IP address called?

Octet ✓ Correct Answer

How many sections (in digits) does an IPv4 address have?

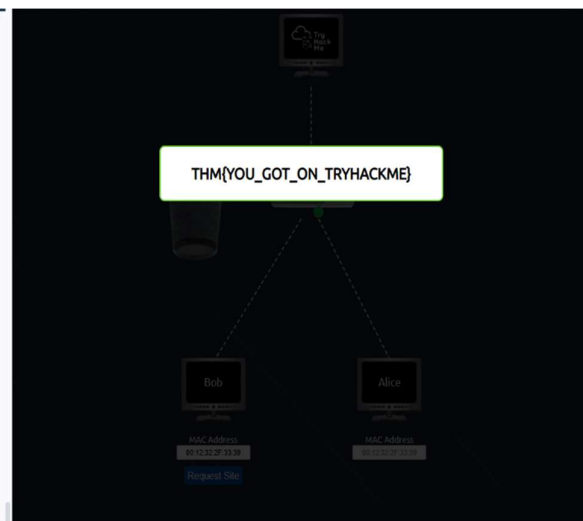
4 ✓ Correct Answer [Hint](#)

What does the term "MAC" stand for?

Media Access Control ✓ Correct Answer

Deploy the interactive lab using the "View Site" button and spoof your MAC address to access the site. What is the flag?

THM{YOU_GOT_ON_TRYHACKME} ✓ Correct Answer



Practical Sol task 03:

1. Internet Protocol
2. Octet
3. 4
4. Media Access Control
5. THM{YOU_GOT_ON_TRYHACKME}

Task 04: Ping (ICMP)

Task 4: Ping (ICMP)

Ping is one of the most fundamental network tools available to us. Ping uses **ICMP** (Internet Control Message Protocol) packets to determine the performance of a connection between devices. For example, if the connection exists or is reliable.

The time taken for ICMP packets traveling between devices is measured by ping, such as in the screenshot below. This measuring is done using ICMP's echo packet and then ICMP's echo reply from the target device.

Ping can be performed against devices on a network, such as your home network or someone else's website. This tool can be easily used and comes installed on Operating Systems (OSs) such as Linux and Windows. The syntax to do a simple ping is **ping IP_address or_websiteURL**. Let's see this in action in the screenshot below.

```
root@kali:~# ping -c 4 10.10.10.10
PING 10.10.10.10: 56(84) bytes of data:
64 bytes from 10.10.10.10: icmp_seq=1 ttl=64 time=0.13 ms
64 bytes from 10.10.10.10: icmp_seq=2 ttl=64 time=0.12 ms
64 bytes from 10.10.10.10: icmp_seq=3 ttl=64 time=0.14 ms
64 bytes from 10.10.10.10: icmp_seq=4 ttl=64 time=0.13 ms
^C
--- 10.10.10.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 0.000ms
rtt min/avg/max/mdev = 0.132/0.136/0.137/0.005 ms
```

Here we are pinging a device that has the private address of 10.10.10.10. Ping informs us that we have sent our ICMP packets, all of which were received with an average time of 0.13 milliseconds.

Now you are going to do the same thing to ping the address of "8.8.8.8" on the deployable website in this task. Pinging the correct address will reveal a flag to answer the following question below.

Answer the questions below

What protocol does ping use?

ICMP ✓ Correct Answer

What is the syntax for ping 10.10.10.10?

ping 10.10.10.10 ✓ Correct Answer

What flag do you get when you ping 8.8.8.8?

THM{I_PINGED_THE_SERVER} ✓ Correct Answer

IP Address

Send Ping Request

```
user@kali:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=64 time=0.16 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=64 time=0.15 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=64 time=0.15 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=64 time=0.15 ms
^C
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 0.000ms
rtt min/avg/max/mdev = 0.152/0.150/0.151/0.005 ms
```

user@kali:~\$ ping -c 4

Practical Sol task 04:

- 1. ICMP (Internet Control Message Protocol)
- 2. Ping 10.10.10.10
- 3. Flag: THM{I_PINGED_THE_SERVER}

You did it! 🎉 What is Networking? complete!

Points earned

80

Completed tasks

5

Room type

Walkthrough

Difficulty

Info

Streak

2

81,927 users are actively learning this week